Towards infinitely many local extrema of the optimal value $w_{n,p}$ of the all-heads coin game for $p < \frac{1}{2}$

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DEPRECATED. This development note has been superseded by the unified Paper 2 manuscript `Oscillation/manuscript.tex`, which is now the paper of record. It is kept only as a development record — its detailed proofs may assist when migrating content into the manuscript — and is no longer maintained.

Abstract

This note concerns the conjecture that, for $p < \frac{1}{2}$, the optimal winning probability $w_{n,p}$ of the all-heads coin game has infinitely many strict local maxima and minima as a function of n. We reduce the conjecture to two one-sided statements. The first — $w_{n,p}$ *is not eventually non-decreasing* — is proved here unconditionally and elementarily. The second — $w_{n,p}$ *is not eventually non-increasing* — is reduced, via an *exact* transfer lemma, to the log-periodic oscillation of the strategy-All value $b_{n,p}$ that is derived (modulo two standard analytic gaps) in the companion note `strategy_all_asymptotics.tex`. We record carefully which steps are rigorous and which remain open, and isolate the single new analytic point that a complete proof would still require.

1 Setup, standing facts, and the conjecture

Fix $p \in (0,1)$, write $q := 1-p$, and let $w_n := w_{n,p}$ denote the optimal winning probability of the all-heads coin game with n coins. By definition $w_0 = 1$, and for $n \geq 1$ the Bellman equation reads

$$w_n \;=\; p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \max_{j \leq m \leq n-1} w_m, \tag{1}$$

the empty sum at $n = 1$ giving $w_1 = p$.

Standing facts. We use the following throughout, all elementary or established in the main manuscript.

(F1) $0 < w_n \leq 1$ for every n (it is a probability, and $w_n \geq p^n > 0$).

(F2) For $p < \frac{1}{2}$ one has $q > \frac{1}{2} > p$; hence $q > p$, $p/q < 1$, and $2p < 1$.

(F3) $\sum_{j=0}^{n} \binom{n}{j} p^{n-j} q^j = 1$, $\sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j = 1 - p^n - q^n$, and $\sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} q^j = 1 - q^n$.

(F4) The strategy-One and strategy-All values satisfy the *linear* recursions

$$a_n = p^n + (1 - p^n - q^n)\, a_{n-1}, \qquad b_n = \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} q^j\, b_j,$$

with $a_0 = b_0 = 1$, and $w_n \geq \max(a_n, b_n)$ for all n (the optimum dominates any fixed strategy).

The conjecture. For $p > \frac{1}{2}$ the sequence $n \mapsto w_{n,p}$ is strictly increasing (Theorem 3.1 of the main paper), and for $p = \frac{1}{2}$ it is constant. The case $p < \frac{1}{2}$ is the interesting one.

Conjecture 1.1. For every $p \in (0, \frac{1}{2})$ the sequence $n \mapsto w_{n,p}$ is *not eventually monotone*; equivalently, it has infinitely many strict local maxima and infinitely many strict local minima.

The two formulations agree up to the (expected, and in all computed data confirmed) genericity assumption that $w_{n,p} \neq w_{n+1,p}$ for all n; see Lemma 2.1 and Remark 6.3. Conjecture 1.1 is consistent with the numerical extremum data of the main manuscript (for $p = 0.45$, local minima at $n = 6, 17$ and a local maximum at $n = 15$; for $p = 0.42$, minima at $n = 7, 13$ and a maximum at $n = 9$) and with van Doorn's description of the optimal strategy for $p < \frac{1}{2}$ as "infinitely complex".

2 Reduction to two one-sided statements

For a real sequence (x_n) put $d_n := x_{n+1} - x_n$. Call $n \geq 1$ a *local maximum* if $x_{n-1} \leq x_n \geq x_{n+1}$ and a *strict* local maximum if both inequalities are strict; symmetrically for minima. The sequence is *eventually non-decreasing* if $d_n \geq 0$ for all large n, *eventually non-increasing* if $d_n \leq 0$ for all large n, and *eventually monotone* if it is one of the two.

Lemma 2.1. *For any sequence (x_n) the following are equivalent.*

(i) *(x_n) is not eventually monotone.*

(ii) *$d_n > 0$ for infinitely many n* and *$d_n < 0$ for infinitely many n.*

(iii) *(x_n) is neither eventually non-decreasing nor eventually non-increasing.*

Any of these implies that (x_n) has infinitely many local maxima and infinitely many local minima. If moreover $x_n \neq x_{n+1}$ for all large n, the local extrema are strict.

Proof. "Eventually non-decreasing" is the negation of "$d_n < 0$ for infinitely many n", and "eventually non-increasing" the negation of "$d_n > 0$ for infinitely many n". Hence "eventually monotone" = "$\neg(d_n < 0 \text{ i.o.}) \vee \neg(d_n > 0 \text{ i.o.})$", whose negation is exactly (ii); and (iii) is the same statement spelled out. This gives (i) $\iff$ (ii) $\iff$ (iii).

Assume (ii). Pick any n_1 with $d_{n_1} < 0$. By (ii) there is a smallest $m > n_1$ with $d_m > 0$; then $d_{m-1} \leq 0 < d_m$, so $x_{m-1} \geq x_m \leq x_{m+1}$, i.e. m is a local minimum. By (ii) again there is a later index with $d < 0$, and the smallest such yields, in the same way, a local maximum beyond m. Iterating produces infinitely many of each. Strictness under $x_n \neq x_{n+1}$ is immediate, since then all the inequalities above are strict. □

By Lemma 2.1, Conjecture 1.1 is equivalent to the conjunction of

$$\text{(A)} \quad (w_{n,p}) \text{ is not eventually non-decreasing}, \qquad \text{(B)} \quad (w_{n,p}) \text{ is not eventually non-increasing}. \tag{2}$$

Section 3 proves (A) unconditionally. Sections 4–5 reduce (B) to the oscillation of strategy All.

3 Statement (A): $w_{n,p}$ is not eventually non-decreasing

Proposition 3.1. *Let $p \in (0, \frac{1}{2})$. There is no M such that $w_n \geq w_{n-1}$ for all $n > M$. That is, $w_n < w_{n-1}$ for infinitely many n.*

The mechanism is the following. On a stretch where w is non-decreasing, the suffix-maximum in (1) collapses to its *right* endpoint w_{n-1}, so the Bellman equation reduces to the strategy-One recursion (F4). Its one-step increment is $p^n(1 - w_{n-1}) - q^n w_{n-1}$, in which the "lose-at-once" term $q^n w_{n-1}$ (all n coins tails) dominates the "win-at-once" term $p^n(1 - w_{n-1})$ because $q > p$. Hence the value must eventually fall.

Proof. Suppose, for contradiction, that $w_n \geq w_{n-1}$ for all $n > M$, i.e. w is non-decreasing on $\{M, M+1, \dots\}$. Fix $n > M + 1$ and split the Bellman sum (1) at $j = M$:

$$w_n \;=\; p^n \;+\; \underbrace{\sum_{j=1}^{M-1} \binom{n}{j} p^{n-j} q^j \, S_{j,n}}_{=:T_n} \;+\; \sum_{j=M}^{n-1} \binom{n}{j} p^{n-j} q^j \, S_{j,n}, \qquad S_{j,n} := \max_{j \leq m \leq n-1} w_m.$$

For $j \geq M$ the index range $\{j, \dots, n-1\}$ lies in $\{M, M+1, \dots\}$, where w is non-decreasing, so $S_{j,n} = w_{n-1}$. For every j the range contains $n - 1$, hence $w_{n-1} \leq S_{j,n} \leq 1$ by (F1). Writing $U_n := \sum_{j=1}^{M-1} \binom{n}{j} p^{n-j} q^j$ and using (F3),

$$w_n = p^n + T_n + \big(1 - p^n - q^n - U_n\big)\, w_{n-1},$$

and therefore

$$w_n - w_{n-1} \;=\; p^n\,(1 - w_{n-1}) \;-\; q^n\, w_{n-1} \;+\; \big(T_n - U_n w_{n-1}\big). \tag{3}$$

Since $w_{n-1} \leq S_{j,n} \leq 1$, every summand of T_n lies between $\binom{n}{j} p^{n-j} q^j w_{n-1}$ and $\binom{n}{j} p^{n-j} q^j$, so $0 \leq T_n - U_n w_{n-1} \leq U_n$. Moreover, for $1 \leq j \leq M - 1$,

$$\binom{n}{j} p^{n-j} q^j \leq n^{\,M-1}\, p^{n-M+1},$$

because $\binom{n}{j} \leq n^j \leq n^{M-1}$, $q^j \leq 1$, and $p^{n-j} \leq p^{n-M+1}$ (smaller exponent, $0 < p < 1$). Hence

$$U_n \;\leq\; (M-1)\, n^{M-1} p^{n-M+1} \;=\; C_1\, n^{M-1} p^n, \qquad C_1 := (M-1)\, p^{1-M}.$$

Dropping the nonnegative term $p^n(1 - w_{n-1}) \leq p^n$ as an upper bound and using $w_{n-1} \geq w_M =: c > 0$ (non-decreasing, and $w_M > 0$ by (F1)), (3) gives

$$w_n - w_{n-1} \;\leq\; p^n + U_n - q^n w_{n-1} \;\leq\; \big(1 + C_1 n^{M-1}\big)\, p^n \;-\; c\, q^n.$$

The ratio of the two terms is $\big(1 + C_1 n^{M-1}\big) c^{-1} (p/q)^n \to 0$ as $n \to \infty$, since $p/q < 1$ by (F2) kills the polynomial factor. Thus there is n_0 such that $w_n - w_{n-1} < 0$ for all $n \geq n_0$, contradicting $w_n \geq w_{n-1}$ for $n > M$. $\square$

Remark 3.2. Proposition 3.1 already refutes the natural first guess "for every p there is an N with $w_{n,p}$ increasing for $n > N$". It also shows that the main manuscript's first-order Corollary — "$w_{n,p}$ is eventually increasing" — is genuinely a *first-order* statement: the exact sequence is not eventually increasing. The obstruction $q^n w_{n-1}$ is exponentially small in n but, for fixed $p < \frac{1}{2}$, eventually beats every term that a finite δ-expansion around $p = \frac{1}{2}$ can see.

4 An exact transfer lemma for non-increasing tails

We now turn to statement (B). The key is that on a non-increasing stretch the suffix-maximum in (1) collapses to its *left* endpoint, turning the Bellman equation into the *linear* strategy-ALL recursion.

Lemma 4.1 (Transfer to strategy ALL)**.** *Let $p \in (0,1)$ and suppose $w_n \geq w_{n+1}$ for all $n \geq M$ (some $M \geq 1$). Define a sequence $(x_n)_{n\geq 0}$ by*

$$x_0 := 1, \qquad x_j := \max_{j\leq m\leq M} w_m \quad (1 \leq j \leq M-1), \qquad x_j := w_j \quad (j \geq M).$$

*Then $x_n = w_n$ for all $n \geq M$, and (x_n) satisfies the exact strategy-*ALL *recursion*

$$x_n \;=\; \sum_{j=0}^{n-1} \binom{n}{j} p^{\,n-j} q^{\,j}\, x_j \qquad \text{for every } n > M. \tag{4}$$

Proof. Fix $n > M$ and consider $S_{j,n} = \max_{j\leq m\leq n-1} w_m$ in the Bellman equation (1).

- For $M \leq j \leq n-1$: the range $\{j, \dots, n-1\}$ lies in $\{M, M+1, \dots\}$, where w is non-increasing, so the maximum is at the left endpoint, $S_{j,n} = w_j = x_j$.
- For $1 \leq j \leq M-1$: split $\{j, \dots, n-1\} = \{j, \dots, M-1\} \cup \{M, \dots, n-1\}$. On the second block w is non-increasing, so its maximum is w_M; hence $S_{j,n} = \max\bigl(\max_{j\leq m\leq M-1} w_m,\ w_M\bigr) = \max_{j\leq m\leq M} w_m = x_j$, independent of n.

Substituting into (1) and writing the term p^n as $\binom{n}{0}p^n q^0 x_0$ (with $x_0 = 1$) yields $w_n = \sum_{j=0}^{n-1}\binom{n}{j}p^{n-j}q^j x_j$. Since $x_n = w_n$ for $n \geq M$, this is (4). □

Remark 4.2. For $M = 1$ the lemma specialises to the reduction lemma recorded in `CONVERSATION_LOG.md` (and `localmaximum.tex`): if w is non-increasing on all of $\mathbb{N}$, then $w_n = b_n$ for every n. Lemma 4.1 is the version needed when w is only *eventually* non-increasing: w then equals, on its tail, a strategy-ALL sequence with finitely many initial values replaced.

Generating-function form. Let $X(z) := \sum_{n\geq 0} x_n z^n/n!$ be the exponential generating function of the transferred sequence. The binomial convolution underlying (4) has, exactly as in `strategy_all_asymptotics.tex`, the EGF $e^{pz}X(qz)$; subtracting the diagonal term $q^n x_n$ (EGF $X(qz)$) gives that $\sum_{j=0}^{n-1}\binom{n}{j}p^{n-j}q^j x_j$ has EGF $(e^{pz}-1)X(qz)$. Since (4) holds for all $n > M$ but not necessarily for $n \leq M$, the difference $X(z) - (e^{pz}-1)X(qz)$ is a polynomial of degree at most M:

$$X(z) \;=\; P(z) \;+\; \bigl(e^{pz}-1\bigr)\, X(qz), \qquad \deg P \leq M. \tag{5}$$

This is precisely the q-difference equation (1.3) of `strategy_all_asymptotics.tex`, with the constant 1 replaced by the polynomial P (which encodes the finitely many "wrong" initial values $x_1, \dots, x_M$; for the genuine strategy-ALL EGF B one has $P \equiv 1$). Iterating (5), with the remainder $\prod_{i<K}(e^{pq^i z}-1)\, X(q^K z) \to 0$ as $K \to \infty$ because each factor $e^{pq^i z} - 1 \to 0$,

$$X(z) \;=\; \sum_{k\geq 0} \Bigl[\prod_{i=0}^{k-1}\bigl(e^{pq^i z}-1\bigr)\Bigr] P(q^k z). \tag{6}$$

The point of (6) is structural: the large-z behaviour of X is driven by exactly the same product $\prod_i(e^{pq^i z}-1)$ as that of the strategy-ALL EGF B; the polynomial P enters only as a degree-$\leq M$ modulation of the summands. In particular the saddle-point / Mellin analysis of `strategy_all_asymptotics.tex` applies verbatim, and the oscillatory poles — located at $s_\ell = 2\pi i\ell/\log(1/q)$, $\ell \in \mathbb{Z}$, and produced by the geometric factor $1/(1-q^{-s})$ that the q-dilation $z \mapsto qz$ generates — are unchanged. Only the *values* of the Fourier coefficients depend on P.

5 Statement (B), conditional on the oscillation of strategy All

Hypothesis 5.1 (All-oscillation)**.** For every $p \in (0, \frac{1}{2})$ the strategy-All value satisfies $b_{n,p} = \Phi_p(\log n)\,(1+o(1))$ as $n \to \infty$, where Φ_p is continuous, periodic with period $\log(1/q)$, and *non-constant*; equivalently $\liminf_n b_{n,p} < \limsup_n b_{n,p}$.

Hypothesis 5.2 (transferred All-oscillation)**.** For every $p \in (0, \frac{1}{2})$ and every $M \geq 1$, any sequence (x_n) produced by Lemma 4.1 from an eventually non-increasing tail of $(w_{n,p})$ is *not* eventually monotone.

Theorem 5.3 (conditional)**.** *Assume Hypothesis 5.2. Then for every $p \in (0, \frac{1}{2})$ the sequence $(w_{n,p})$ is not eventually non-increasing. Together with Proposition 3.1 this gives Conjecture 1.1: $(w_{n,p})$ is not eventually monotone, hence has infinitely many local maxima and minima.*

Proof. Suppose $(w_{n,p})$ were non-increasing on $\{M, M+1, \dots\}$. By Lemma 4.1 the transferred sequence (x_n) satisfies $x_n = w_n$ for $n \geq M$; in particular (x_n) is non-increasing for $n \geq M$, hence *eventually monotone*. This contradicts Hypothesis 5.2. So $(w_{n,p})$ is not eventually non-increasing, which is statement (B) of (2). Combining with statement (A) (Proposition 3.1) and Lemma 2.1 proves the theorem. □

So the entire conjecture rests on Hypothesis 5.2. We now record what is known about it.

Status of Hypothesis 5.1. The companion note `strategy_all_asymptotics.tex` establishes:

- *Rigorously*: the functional equation $B(z) = 1 + (e^{pz} - 1)B(qz)$, the iterated series, the surjection formula, and the probabilistic interpretation of $b_{n,p}$.
- *By a saddle-point/Mellin derivation that is not yet a full proof*: the log-periodic asymptotics $b_{n,p} = \Phi_p(\log n)(1+o(1))$ with Fourier expansion $b_{n,p} = \sum_\ell \gamma_\ell(p)\, n^{2\pi i\ell/\log(1/q)}(1+o(1))$. Two analytic gaps remain, both "standard in spirit" (Flajolet–Sedgewick): (a) Hayman-admissibility of the entire function B — which is of order 1 and type 1 (indeed $B(r) \leq e^r$, since $0 \leq b_{n,p} \leq 1$), the delicate point being the non-constant log-periodic factor rather than the growth rate; (b) justification of the Mellin contour shift, i.e. the strip of analyticity.
- *Numerically*: the predicted oscillation amplitude is confirmed across fourteen orders of magnitude in p.

A pleasant point is that, *granted* the Mellin framework, the non-constancy of Φ_p is not a separate miracle. The note computes the Fourier coefficients as

$$c_\ell \;=\; -\frac{1}{\log(1/q)}\, p^{-s_\ell}\, \Gamma(s_\ell)\, \zeta(s_\ell + 1), \qquad s_\ell = \frac{2\pi i\ell}{\log(1/q)}.$$

For $\ell \neq 0$ the argument s_ℓ is purely imaginary and nonzero; Γ has no zeros, $p^{-s_\ell} \neq 0$, and $\zeta(s_\ell+1) = \zeta(1+it) \neq 0$ because the Riemann zeta function has no zeros on the line $\Re s = 1$ (the same non-vanishing that underlies the prime number theorem). Hence $c_\ell \neq 0$ for all $\ell \neq 0$: *within the Mellin framework, the oscillation of $b_{n,p}$ is provably genuine.* Thus Hypothesis 5.1 is "true modulo gaps (a)–(b)".

From Hypothesis 5.1 to Hypothesis 5.2. What Theorem 5.3 actually needs is the slightly stronger Hypothesis 5.2, about solutions of the P-*modified* equation (5). By (6) such an X has the same dominant product structure and the same oscillatory poles s_ℓ as B; the Mellin analysis yields a Fourier expansion

$$x_n = \sum_{\ell \in \mathbb{Z}} \gamma_\ell^X(p)\, n^{2\pi i \ell/\log(1/q)}\,(1+o(1)),$$

with the *same* period $\log(1/q)$ and with coefficients γ_ℓ^X that differ from γ_ℓ only through a P-dependent factor (the Mellin transform of the P-part). A non-constant such expansion is not eventually monotone, because a non-constant continuous periodic function attains its maximum and minimum on every period. Hence Hypothesis 5.2 holds *provided* the modulation by P does not annihilate the $\ell = \pm 1$ coefficient.

6 The remaining gap, and why the two sides are asymmetric

Assembling Sections 3–5, a complete proof of Conjecture 1.1 is reduced to a finite, well-delimited list:

(G1) *Hayman-admissibility* of the entire function $B(z)$ (companion-note gap (a)).

(G2) *Mellin contour justification*: the strip of analyticity and the contour shift (companion-note gap (b)).

(G3) *Non-vanishing under modulation*: the P-dependent factor multiplying the $\ell = \pm 1$ pole-residue does not vanish for the specific polynomial P that arises, via Lemma 4.1 and (5), from an eventually non-increasing tail of $(w_{n,p})$.

Gaps (G1)–(G2) are inherited from the companion note and are technical rather than conceptual. Gap (G3) is the one genuinely new analytic point introduced here. It cannot be waved away: P is not free, but it is also not the constant 1, and one must exclude an accidental cancellation. We expect no cancellation — it would be a non-generic coincidence, and $\gamma_{\pm 1}$ already carries the structurally non-zero factor $\Gamma(s_{\pm 1})\zeta(1 \pm it)$ — but a proof requires either an explicit computation of the P-factor or a softer argument (for instance, showing that a convergent solution of (4) would force constraints on $x_0, \dots, x_M$ incompatible with $x_n = w_{n,p} \geq b_{n,p}$). Section 7 carries out the first steps of that softer argument: it removes the dependence on the specific P and reduces gap (G3) to a single, universal non-degeneracy.

Remark 6.1 (why (A) is elementary but (B) is not)**.** The asymmetry between the two halves is structural. On an *increasing* tail the suffix-maximum collapses to the *right* endpoint w_{n-1}, turning (1) into the strategy-One recursion: a contractive linear recursion whose one-step increment has the *sign-definite* dominant term $-q^n w_{n-1}$. That is what makes Proposition 3.1 elementary. On a *decreasing* tail the suffix-maximum collapses to the *left* endpoint w_j, turning (1) into the strategy-All recursion: a q-dilation-driven linear recursion whose solutions are log-periodic, with *no* sign-definite drift. Ruling out monotonicity there is exactly "a non-constant periodic function is not monotone", and the non-constancy is the analytic content carried by $\zeta(1+it) \neq 0$. There is, correspondingly, no elementary contradiction available for case (B): the crude estimates (weights concentrating at $j \approx qn$, etc.) are all consistent with a monotone convergent tail; only the finer Fourier structure breaks it.

Remark 6.2 (a weaker, more accessible statement)**.** A weaker companion to Conjecture 1.1 — that $(w_{n,p})$ has *at least one* strict local maximum, i.e. is not globally non-increasing — follows from the $M = 1$ case of Lemma 4.1: if w were non-increasing on all of $\mathbb{N}$ then $w_n = b_n$ for every n, so it would suffice that $b_{n,p}$ is not monotone non-increasing. For specific p this is rigorous by exact rational computation (e.g. $p = \frac{21}{50}$, where $w_9 > w_8$ and $w_9 > w_{10}$; see the main manuscript and `Manuscript/proof_intermediate_p042.tex`). A uniform statement over all $p \in (0, \frac{1}{2})$ again needs Hypothesis 5.1.

Remark 6.3. Lemma 2.1 delivers *strict* extrema once $w_n \neq w_{n+1}$ for all large n. For $p < \frac{1}{2}$ this is expected and holds in all computed data; a clean sufficient condition would be that $w_{n,p}$, a rational function of p of growing degree, never has two equal consecutive values — plausibly provable by a degree/positivity argument but not pursued here. Absent it, Conjecture 1.1 still holds with "strict" replaced by the non-strict notion of local extremum.

7 Progress on gap (G3): removing the dependence on P

Throughout this section we work under the hypothesis to be refuted: $(w_{n,p})$ is non-increasing on $\{M, M+1, \dots\}$. By Lemma 4.1 the transferred sequence satisfies $x_n = w_n$ for $n \geq M$; being non-increasing and bounded it converges,

$$x_n = w_n \longrightarrow L \qquad (n \to \infty), \qquad L \geq 0.$$

Gap (G3) asks us to contradict this. We show that the obstruction can be detached from the specific polynomial P.

7.1 An elementary comparison: the sequence $g = w - b$

Lemma 7.1. *Put $g_n := x_n - b_n$, with b the strategy-*ALL *value of (F4). Then*

(i) *$g_0 = 0$, and $g_n \geq 0$ for every $n \geq 0$;*

(ii) *g satisfies the homogeneous strategy-*ALL *recursion $g_n = \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} q^j g_j$ for every $n > M$;*

(iii) *the running maximum is non-increasing: $g_n < \max_{0 \leq j < n} g_j$ for every $n > M$, so $\max_{0 \leq j \leq n} g_j = \max_{0 \leq j \leq M} g_j =: G^*$ for all $n \geq M$; in particular $0 \leq g_n \leq G^*$ for all n.*

Proof. (i) $g_0 = x_0 - b_0 = 1 - 1 = 0$. For $j \geq M$, $g_j = w_j - b_j \geq 0$ since the optimum dominates strategy ALL ((F4)). For $1 \leq j \leq M-1$, Lemma 4.1 gives $x_j = \max_{j \leq m \leq M} w_m \geq w_j \geq b_j$, so $g_j \geq 0$.

(ii) Both x and b satisfy $y_n = \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} q^j y_j$ for $n > M$ — x by Lemma 4.1, b for every $n \geq 1$. Subtract.

(iii) Fix $n > M$. By (i) all $g_j \geq 0$, so by (ii) and (F3),

$$g_n = \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} q^j g_j \leq \Big(\max_{0 \leq j < n} g_j \Big) \sum_{j=0}^{n-1} \binom{n}{j} p^{n-j} q^j = (1 - q^n) \max_{0 \leq j < n} g_j < \max_{0 \leq j < n} g_j.$$

Hence the running maximum never increases past $n = M$. □

Proposition 7.2. *Under the standing hypothesis, exactly one of the following holds.*

(a) *$g_n = 0$ for all $n \geq M$. Then $w_n = b_n$ for all $n \geq M$; since $b_{n,p}$ is not eventually monotone (Hypothesis 5.1), neither is $(w_{n,p})$, contradicting the standing hypothesis.*

(b) *$g_n > 0$ for some $n \geq M$. Then g is a non-zero, non-negative, bounded solution of the homogeneous strategy-*ALL *recursion (for $n > M$) with $g_0 = 0$.*

Hence gap (G3) is equivalent to excluding case (b).

The $g \equiv 0$ branch is thus closed rigorously and elementarily; it uses only Hypothesis 5.1. In case (b), g would — since $x_n \to L$ and b_n oscillates — have to oscillate in exact anti-phase with b, with asymptotic profile $L - \Phi$. Non-negativity alone does not forbid this: by Lemma 7.1(iii) it only yields $0 \leq g_n \leq G^*$, and a non-negative log-periodic function is admissible. The oscillation must be defeated through its Fourier structure.

7.2 Auxiliary functions B_r and the elimination of P

Write the modulating polynomial of (5) as $P(z) = \sum_{r=0}^{M} a_r z^r$, and set $\phi_k(z) := \prod_{i=0}^{k-1}(e^{pq^i z} - 1)$, so that the iterated series (6) is $X(z) = \sum_{k\geq 0} \phi_k(z) P(q^k z)$. Expanding $P(q^k z) = \sum_r a_r q^{rk} z^r$ and exchanging the sums,

$$X(z) = \sum_{r=0}^{M} a_r\, z^r\, B_r(z), \qquad B_r(z) := \sum_{k\geq 0} q^{rk} \phi_k(z). \tag{7}$$

Lemma 7.3. *Each B_r is entire and satisfies $B_r(z) = 1 + q^r(e^{pz} - 1)B_r(qz)$, with $B_0 = B$.*

Proof. $\phi_0 \equiv 1$ and $\phi_k(z) = (e^{pz} - 1)\phi_{k-1}(qz)$, so $B_r(z) = 1 + \sum_{k\geq 1} q^{rk}(e^{pz} - 1)\phi_{k-1}(qz) = 1 + q^r(e^{pz} - 1)\sum_{k\geq 0} q^{rk}\phi_k(qz)$. Convergence and entirety are as for B in the companion note. ☐

Identity (7) is the decisive bookkeeping step: it expresses the transferred EGF as a fixed linear combination, with the coefficients a_r of P, of the P-*independent* functions $z^r B_r(z)$.

7.3 Reduction of (G3) to a P-independent non-degeneracy

At the level of the saddle-point analysis of the companion note — hence modulo gaps (G1)–(G2) — each B_r has a log-periodic profile. In $B_r(z) = \sum_k q^{rk}\phi_k(z)$ the dominant index is $k \approx K(z) = \log(pz)/\log(1/q)$, where $q^{rK(z)} = (pz)^{-r}$ and the saddle is $O(1)$ wide; thus

$$B_r(z) \;\sim\; (pz)^{-r}\, e^z\, \Psi_r(\log z), \tag{8}$$

with Ψ_r continuous, periodic of period $\log(1/q)$, and $\Psi_0 = \Phi$. Substituting (8) into (7), the factors z^r cancel $(pz)^{-r}$ up to p^{-r}:

$$X(z) \;\sim\; e^z\, \Psi^X(\log z), \qquad \Psi^X := \sum_{r=0}^{M} a_r\, p^{-r}\, \Psi_r. \tag{9}$$

The transferred sequence converges if and only if its profile Ψ^X is constant.

Proposition 7.4. *Assume the asymptotics* (8)*. If $\Psi_0, \ldots, \Psi_M$ are linearly independent modulo the constants, then the only convergent solution of the transferred recursion is the zero sequence. As $x_0 = 1$, the sequence x does not converge, and gap (G3) is closed.*

Proof. If Ψ^X is constant then, by independence, $a_r p^{-r} = 0$ for every r, i.e. $P \equiv 0$. But $P \equiv 0$ in (5) forces, on iterating, $X(z) = \phi_K(z)X(q^K z) \to 0$, so $X \equiv 0$ and $x \equiv 0$, against $x_0 = 1$. Hence Ψ^X is non-constant, and a non-constant continuous periodic function is not eventually monotone. ☐

The hypothesis of Proposition 7.4 is genuinely P-independent: it speaks only of the universal family $(\Psi_r)_{r\geq 0}$. That this is the *right* reformulation, with no loss, is confirmed by an exact observation. If $\sum_{r=0}^{M} c_r\Psi_r$ were constant, set $Y(z) := \sum_{r=0}^{M} c_r z^r B_r(z)$; using $B_r(z) = 1 + q^r(e^{pz} - 1)B_r(qz)$ and $z^r = q^{-r}(qz)^r$,

$$Y(z) = \widetilde{C}(z) + (e^{pz} - 1)\,Y(qz), \qquad \widetilde{C}(z) := \sum_{r=0}^{M} c_r z^r,$$

the *same* q-difference equation (5). Hence gap (G3), for every M, is equivalent to

$(\Psi_r)_{r\geq 0}$ linearly independent modulo constants $\iff$ no non-zero polynomial $\widetilde{C}$ yields a convergent ALL-type seq

7.4 Status of (G3) after this reduction

- **Gained.** Gap (G3) no longer refers to the polynomial P of the specific tail of w. By Proposition 7.4 it is reduced to one universal non-degeneracy: linear independence, modulo constants, of the profiles Ψ_r of the auxiliary functions $B_r = 1 + q^r(e^{pz} - 1)B_r(qz)$. The $g \equiv 0$ branch (Proposition 7.2(a)) is moreover fully rigorous.
- **Still open.** The independence of the Ψ_r. Via the Mellin formalism it is the assertion that the Fourier-coefficient matrix $[\widehat{\Psi}_{r,\ell}]_{r\geq 0,\,\ell\in\mathbb{Z}}$ has full row rank. We expect this — for $r = 0$ the relevant coefficients already carry the structurally non-zero factor $\Gamma(s_\ell)\zeta(s_\ell + 1)$ — but it is not proved, and it inherits gaps (G1)–(G2).
- **What does not suffice.** Non-negativity of g (Lemma 7.1): the running-maximum bound gives only $0 \leq g_n \leq G^*$, compatible with a non-negative oscillation of profile $L - \Phi$.

In short, gap (G3) is now sharp and P-free: it asserts that no non-trivial polynomial modulation of the strategy-ALL q-difference equation produces a convergent sequence.

8 A renormalisation route to Hypothesis 5.2

Sections 5–7 reduce Hypothesis 5.2 to the Mellin analysis of the companion note. This section sketches an alternative, more self-contained route: a scale-by-scale induction that replaces the Mellin/Hayman machinery (gaps (G1)–(G2)) by a single elementary *propagation estimate*. It does not remove the non-degeneracy (G3), but it isolates it as one finite, numerically accessible base case.

Throughout, (x_n) denotes a bounded solution of the strategy-ALL recursion for $n > M$ — a transferred sequence of Lemma 4.1, or b itself.

8.1 The recursion as a smoothing in log-scale

Lemma 8.1. *For $n > M$, with $J_n \sim \mathrm{Bin}(n, q)$,*

$$x_n = \frac{E[x_{J_n}]}{1 + q^n}.$$

Proof. Since $\binom{n}{j}p^{n-j}q^j = P(J_n = j)$, $\sum_{j=0}^{n-1}\binom{n}{j}p^{n-j}q^j x_j = E[x_{J_n}] - P(J_n = n)x_n = E[x_{J_n}] - q^n x_n$. Equate with x_n and solve. $\square$

As $E[J_n] = qn$ and $\mathrm{Var}(J_n) = npq$, the index J_n concentrates at qn with relative spread $\sqrt{p/(qn)} \to 0$. Pass to the logarithmic variable: put

$$t := \log n, \qquad L := \log(1/q), \qquad \omega := 2\pi/L,$$

and let y be the log-linear interpolation of $n \mapsto x_n$, so $y(\log n) = x_n$. The decisive arithmetic fact is

$$\log(qn) = \log n - L :$$

one application of the recursion shifts the log-scale down by *exactly one period.* By the delta method $\log J_n$ has mean $t - L + O(1/n)$ and standard deviation $\sigma(t) = \sqrt{p/q}\, e^{-t/2}\,(1 + o(1)) \to 0$. Hence Lemma 8.1 reads, up to the negligible factor $1/(1 + q^n) = 1 - O(e^{-Le^t})$,

$$y(t) = E\big[y\big(t - L + \sigma(t)\, Z_t\big)\big] + \varepsilon(t), \tag{10}$$

where Z_t is the standardised $\log J_n$ and $\varepsilon(t)$ collects the $1/(1 + q^n)$ factor and the indices $j \le M$ (total binomial weight $O(n^M p^n)$), so $|\varepsilon(t)| = O(e^{-c\,e^t})$. In words: *$y$ at log-scale t equals y at log-scale $t - L$, smoothed by a kernel of width $\sigma(t) \to 0$.*

8.2 The local fundamental coefficient

Fix a window $\chi \in C_c^\infty(\mathbb{R})$ with

$$\int_{\mathbb{R}} \chi = 1, \qquad \hat{\chi}(2\pi) := \int_{\mathbb{R}} \chi(u)\, e^{-2\pi i u}\, du = 0$$

(one linear constraint, easily satisfied). Define the *local fundamental coefficient* of y at log-scale t:

$$A(t) := \frac{1}{L} \int_{\mathbb{R}} y(s)\, e^{-i\omega s}\, \chi\Big(\tfrac{s-t}{L}\Big)\, ds. \tag{11}$$

Lemma 8.2. *If $x_n \to x_\infty$, then $A(t) \to 0$ as $t \to \infty$.*

Proof. For large t the window sees only s with $y(s) = x_\infty + o(1)$. The leading term is $\frac{x_\infty}{L} \int e^{-i\omega s} \chi(\frac{s-t}{L})\, ds = x_\infty e^{-i\omega t} \hat{\chi}(\omega L) = x_\infty e^{-i\omega t} \hat{\chi}(2\pi) = 0$ since $\omega L = 2\pi$; the $o(1)$ part contributes $o(1)$. $\square$

Thus a non-vanishing A certifies that (x_n) does *not* converge, hence — being bounded — is not eventually monotone.

8.3 The propagation estimate

Proposition 8.3 (Propagation)**.** *There exist $\kappa(t) \in \mathbb{C}$ and a remainder $r(t)$ with*

$$A(t + L) = \kappa(t)\, A(t) + r(t), \qquad |\kappa(t) - 1| = O(e^{-t}), \qquad |r(t)| = O(e^{-t/2}), \tag{12}$$

the implied constants depending only on p, χ and $\sup_n |x_n|$.

Proof sketch. Insert (10) into (11) evaluated at $t + L$. The substitution $s \mapsto s + L$ turns the period shift into the identity on the phase, because $e^{-i\omega L} = e^{-2\pi i} = 1$; removing the smoothing variable $s \mapsto s + \sigma(t) Z_t$ then produces the multiplier

$$\kappa(t) = E\big[e^{i\omega\sigma(t) Z_t}\big] = 1 - \tfrac{1}{2}\omega^2 \sigma(t)^2 + O(\sigma(t)^3) = 1 - O(e^{-t}),$$

the characteristic function of the smoothing kernel at the fundamental frequency. The remainder $r(t)$ collects: (i) the displacement of the window argument by $\sigma(t)Z_t$, of order $O(\sigma(t)) = O(e^{-t/2})$; (ii) the Edgeworth correction to the Gaussian approximation of $\log J_n$, $O(\sigma(t)^3)$; (iii) the interpolation and Euler–Maclaurin errors, $O(1/n) = O(e^{-t})$; (iv) $\varepsilon(t)$, super-exponentially small. The dominant term is (i), giving $|r(t)| = O(e^{-t/2})$. Quantitative forms of (i)–(iv) are standard (Berry–Esseen / Edgeworth bounds for the binomial, uniform in the relevant range of t); the constants are not tracked here. $\square$

The two exponents in (12) are what matters: both $\sum_k |\kappa(t_0 + kL) - 1|$ and $\sum_k |r(t_0 + kL)|$ converge, being geometric series in $e^{-L} < 1$.

The factor κ admits a fully rigorous treatment; only the remainder $r(t)$ rests on the sketch. With $t = \log n$, the recursion of Lemma 8.1 shifts the log-scale down by exactly one period to $\log(qn)$, and κ is the resulting characteristic-function-type average over the binomial law.

Lemma 8.4 (the propagation factor)**.** *Let $0 < q < 1$, $p = 1 - q$, $\omega \in \mathbb{R}$, and $J_n \sim \mathrm{Bin}(n, q)$. Then*

$$\kappa(n) := E\big[(J_n/(qn))^{\,i\omega}\,\mathbf{1}_{\{J_n \ge 1\}}\big]$$

satisfies, for every $n \ge 1$,

$$\kappa(n) = 1 - \frac{p}{2qn}\,(\omega^2 + i\omega) + R_n, \qquad |R_n| \le \frac{C(\omega, q)}{n^{3/2}},$$

with $C(\omega, q)$ explicit. In particular $|\kappa(n) - 1| = O(1/n)$, and $n\,|\kappa(n) - 1| \to \frac{p}{2q}\,|\omega^2 + i\omega| = \frac{p\,\omega\sqrt{\omega^2+1}}{2q}$.

Proof. Put $V_n := J_n/(qn) - 1$, so $E[V_n] = 0$ and $-1 \le V_n \le 1/q$. The central moments of $\mathrm{Bin}(n, q)$ give the exact values $E[V_n^2] = \mathrm{Var}(J_n)/(qn)^2 = p/(qn)$ and $E[V_n^4] = 3p^2/(q^2n^2) + p(1-6pq)/(q^3n^3)$; hence, for $n \ge 1/(pq)$, $E[V_n^4] \le 4p^2/(q^2n^2)$ and, by Cauchy–Schwarz,

$$E[|V_n|^3] \le \big(E[V_n^2]\,E[V_n^4]\big)^{1/2} \le 2\,(p/q)^{3/2}\,n^{-3/2}. \tag{13}$$

Taylor expansion. The map $g(v) := (1 + v)^{i\omega} = \exp(i\omega\log(1 + v))$ has $g(0) = 1$, $g'(0) = i\omega$, $g''(0) = i\omega(i\omega - 1) = -(\omega^2 + i\omega)$, and $|g'''(v)| = |\omega|\sqrt{\omega^2 + 1}\,\sqrt{\omega^2 + 4}\,(1 + v)^{-3}$. For $|v| \le \frac{1}{2}$ one has $(1 + v)^{-3} \le 8$, so Taylor's theorem with integral remainder gives

$$\big|(1 + v)^{i\omega} - 1 - i\omega v + \tfrac{1}{2}(\omega^2 + i\omega)v^2\big| \le \tfrac{M_\omega}{6}|v|^3, \quad M_\omega := 8|\omega|\sqrt{\omega^2 + 1}\,\sqrt{\omega^2 + 4}. \tag{14}$$

The tail event. By Hoeffding's inequality $P(|V_n| > \frac{1}{2}) = P(|J_n - qn| > qn/2) \le 2e^{-q^2n/2}$. On $\{|V_n| \le \frac{1}{2}\}$ one has $J_n \ge qn/2 \ge 1$ once $n \ge 2/q$; on the complement $|(1+V_n)^{i\omega}| = 1$ and $|V_n| \le 1/q$.

Assembling (for $n \ge \max(2/q, 1/(pq))$). Split $\kappa(n) = E[(1+V_n)^{i\omega}\mathbf{1}_{|V_n|\le 1/2}] + E[(1+V_n)^{i\omega}\mathbf{1}_{|V_n|>1/2,\,J_n\ge 1}]$; the second term has modulus $\le P(|V_n| > \frac{1}{2}) \le 2e^{-q^2n/2}$. Inserting (14) into the first term,

$$E[(1 + V_n)^{i\omega}\mathbf{1}_{|V_n|\le 1/2}] = P(|V_n| \le \tfrac{1}{2}) + i\omega\,E[V_n\mathbf{1}_{|V_n|\le 1/2}] - \tfrac{1}{2}(\omega^2 + i\omega)\,E[V_n^2\mathbf{1}_{|V_n|\le 1/2}] + \mathcal{E}_n,$$

with $|\mathcal{E}_n| \le \frac{M_\omega}{6}E[|V_n|^3]$. Each truncated moment differs from the full moment ($E[V_n] = 0$, $E[V_n^2] = p/(qn)$) by at most $q^{-2}P(|V_n| > \frac{1}{2}) \le 2q^{-2}e^{-q^2n/2}$, since $|V_n| \le 1/q$. Collecting the exponentially small terms and using (13),

$$\kappa(n) = 1 - \tfrac{p}{2qn}(\omega^2 + i\omega) + R_n, \qquad |R_n| \le \tfrac{M_\omega}{3}(p/q)^{3/2}n^{-3/2} + c(\omega, q)\,e^{-q^2n/2}.$$

As $e^{-q^2n/2} = O(n^{-3/2})$ this is the claim; the finitely many $n < \max(2/q, 1/(pq))$ are absorbed by enlarging $C(\omega, q)$. The limit statement follows since $|R_n| = o(1/n)$. $\square$

Remark 8.5. Lemma 8.4 makes the κ-half of Proposition 8.3 rigorous, and sharper than the sketch: beyond $|\kappa - 1| = O(e^{-t})$ it gives the exact leading constant, including the imaginary part $\frac{p\omega}{2qn}$ produced by the convexity of log. Only the remainder $r(t)$ — the windowed-coefficient bookkeeping — still rests on the proof sketch.

8.4 The induction

Theorem 8.6 (renormalisation form)**.** *Let (x_n) be a bounded solution of the strategy-*All *recursion for $n > M$, and assume the propagation estimate* (12)*. Then for each fixed t_0 the sequence $A(t_0+kL)$ converges as $k \to \infty$ to a limit A_∞. Moreover*

(i) *if x_n converges, then $A_\infty = 0$;*

(ii) *if $A_\infty \neq 0$, then (x_n) is not eventually monotone.*

Proof. Write $A_k := A(t_0 + kL)$, $\kappa_k := \kappa(t_0 + kL)$, $r_k := r(t_0 + kL)$. By (12), $\sum_k |\kappa_k - 1| < \infty$ and $\sum_k |r_k| < \infty$, so the products $\prod_j \kappa_j$ converge with non-zero value. Solving the linear recursion $A_{k+1} = \kappa_k A_k + r_k$,

$$A_k = \Big(\prod_{j<k} \kappa_j\Big) A_0 + \sum_{j<k} \Big(\prod_{j<i<k} \kappa_i\Big) r_j \xrightarrow[k\to\infty]{} A_\infty := \Big(\prod_{j\geq 0} \kappa_j\Big) A_0 + \sum_{j\geq 0} \Big(\prod_{i>j} \kappa_i\Big) r_j,$$

the series converging absolutely; this proves convergence. (i) If $x_n \to x_\infty$ then $A(t) \to 0$ by Lemma 8.2, so $A_\infty = 0$. (ii) If $A_\infty \neq 0$ then $A_k \not\to 0$, so by (i) x_n does not converge; a bounded non-convergent sequence is not eventually monotone. □

So, granted the propagation estimate, Hypothesis 5.2 for a given transferred sequence x is *equivalent* to the single non-degeneracy $A_\infty(x) \neq 0$.

8.5 What the route achieves

- **Replaces (G1)–(G2).** The Mellin contour shift and Hayman-admissibility are replaced by the single Proposition 8.3, whose error terms (i)–(iv) are explicit and elementary; they are summable along the geometric scale precisely because $L = \log(1/q) > 0$. The Mellin transform diagonalises the q-dilation in one stroke; the induction does the same diagonalisation approximately, scale by scale.

- **Isolates (G3).** By linearity A_∞ depends linearly on the data; for a transferred $x = b + g$ (notation of §7) one has $A_\infty(x) = A_\infty(b) + A_\infty(g)$. Here $A_\infty(b) \neq 0$ is the genuine strategy-All oscillation — it coincides with the fundamental Mellin coefficient $\gamma_1(p)$, non-zero by $\Gamma(s_1)\zeta(s_1 + 1) \neq 0$ — and ruling out a cancellation by $A_\infty(g)$ is exactly gap (G3).

- **Makes the base case finite and computable.** Unlike the asymptotic gaps (G1)–(G2), the surviving quantity A_∞ is the limit of the *explicitly computable* sequence $A(t_0 + kL)$ of (11); it is therefore numerically accessible. For the genuine b (Hypothesis 5.1) the route is complete modulo Proposition 8.3 and the single inequality $A_\infty(b) \neq 0$. This picture is borne out numerically: the script `simulation/renorm_validation.py` computes, for $p \in \{0.35, 0.40, 0.42, 0.45\}$, both the windowed coefficient $A(t_0 + kL)$ — found to converge to a non-zero limit — and the factor $\kappa(n)$, for which $n\,|\kappa(n) - 1|$ approaches the constant $\frac{p}{2q}|\omega^2 + i\omega|$ of Lemma 8.4.

9 Summary

- **Rigorous (this note).** Conjecture 1.1 is equivalent to the conjunction of (A) and (B) in (2) (Lemma 2.1). Statement (A), "$w_{n,p}$ is not eventually non-decreasing", is proved unconditionally (Proposition 3.1). The transfer lemma (Lemma 4.1) is exact: on an eventually non-increasing tail, w *is* a strategy-All sequence with finitely many initial values modified.

- **Conditional.** Statement (B), and hence the full conjecture, follows from Hypothesis 5.2 (Theorem 5.3) — the log-periodic oscillation of the P-modified strategy-All recursion.

- **Open.** Hypothesis 5.2 reduces to the gaps of Section 6: the two inherited, technical Mellin/Hayman gaps (G1)–(G2), and gap (G3). The latter is sharpened in Section 7: its $g \equiv 0$ branch is closed rigorously (Proposition 7.2(a)), and the remainder is reduced to a single P-independent non-degeneracy — linear independence, modulo constants, of the profiles of the auxiliary functions B_r (Proposition 7.4). None is a deep obstruction; together they are a concrete finishing project.

- **Alternative route.** Section 8 recasts Hypothesis 5.2 as a scale-by-scale induction: it replaces the Mellin gaps (G1)–(G2) by one elementary propagation estimate (Proposition 8.3) with summable errors, and reduces the conjecture to the single, numerically accessible non-degeneracy $A_\infty \neq 0$.

Suggested use in the main manuscript. Section 6 of the main paper could carry, honestly:

- a *Proposition* — "for $p < \frac{1}{2}$, $w_{n,p}$ is not eventually monotone increasing" — with the short proof of Proposition 3.1; and

- a *Conjecture* — Conjecture 1.1 — with the transfer lemma as the indication of proof and a pointer to `strategy_all_asymptotics.tex` for the analytic input.

This pairs a clean new theorem with a sharply-delimited open problem, and corrects the impression, left by the first-order Corollary of Section 4, that $w_{n,p}$ is eventually increasing.